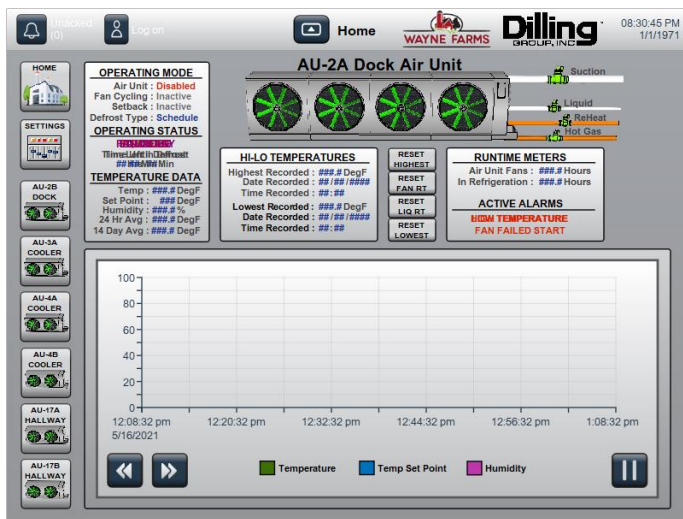


Evaporative Air Units Control

Detail Screen

From the main screen each air unit Detail Screens can be selected by pressing the desired air unit button on the left side of the screen. From this screen you can see detailed information about an air unit including, 1-hour trend, Valve status, Fan status, Runtime meters for fans and for unit cooling. Operating Mode information, Temperature Data including 24 hour & 14 day running average, Time until next defrost, High/Low Temperature, time and date of highest and lowest values recorded. Runtime hours and High/Low temperature data can be reset once logged in.



Evaporator Detail Screen

Automatic Control

Each evaporator is controlled from a temperature/humidity sensor located in the air return side of each unit. A temperature set point for each unit is entered at the Graphics Interface Computer. The system controller will add one (1) and subtract one (1) from the set point that is entered to give a two (2) degree dead band. The reheat solenoid operates from a humidity setpoint and will remain on until the humidity has decreased by 2%. If the room humidity becomes greater than setpoint the reheat will come on "energize" until it reaches the offset. There are four (4) solenoid valves for each evaporator, individually controlled depending on the mode of the evaporators as described below. The units fan starters are controlled together with one (1) output from the controller at the Engine Room MCC.

A Satisfied temperature condition is true when the return air temperature for an evaporator is below the two (2) degree dead band. The units will remain Satisfied until the temperature has risen above the two (2) degree dead band.

Controller Action for Satisfied Units:

- Fans On or may cycle if Fan Cycling is enabled
Fan Cycling should be disabled for Humidity Control.
- Liquid Valve Closed (de-energized).
- Suction Valve Open (energized).
- Hot Gas Valve Closed (de-energized).
- Re-Heat Valve, controlled by Humidity setpoint.

A Call for Cooling condition is true when the return air temperature is above the two (2) degree dead band. The units will continue to cool until the temperature falls below the two (2) degree dead band. The two-degree dead band is plus and minus one from the setpoint. The Unit fans must be running for the Liquid valve to be energized.

Controller Action for Units Cooling:

- Fans On, will override Fan Cycling.
- Liquid Valve Open (energized).
- Suction Valve Open (energized).
- Hot Gas Valve Closed (de-energized).
- Re-Heat Valve, controlled by Humidity setpoint.

Defrost Control

The Evaporators have Hot Gas to defrost the coils. The following is the defrost sequence and controller action which each of the units will go through when defrost is initiated:

Pump Out Cycle:

Pumpout will remove any existing ammonia from the coil and is typically set from 5 minutes to 30 minutes

- Fans On
- Liquid Valve Closed (de-energized).
- Suction Valve Open (energized).
- Hot Gas Valve Closed (de-energized)
- Re-Heat Valve Closed (de-energized) If unit is calling for Re-Heat.

Hot Gas Cycle:

Removes any frost/ice that has formed on the coil and is typically set from 10 minutes to 30 minutes.

- Fans Off
- Liquid Valve Closed (de-energized).
- Suction Valve Closed (de-energized).
- Hot Gas Valve Open (energized)
- Re-Heat Valve Closed (de-energized) If unit is calling for Re-Heat.

Equalize/Bleed Cycle:

Brings the coil back down from Discharge (hot gas) pressure to suction pressure and is typically set from 3 minutes to 10 minutes.

- Fans Off
- HS Liquid Valve Closed (de-energized).
- HS Suction Valve Closed (de-energized).
- Hot Gas Valve Closed (de-energized)
- Re-Heat Valve Closed (de-energized) If unit is calling for Re-Heat.

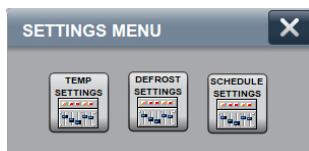
Fan Delay Cycle Mode:

Chills the coil back down and freezes any water droplets on the coil before turning the fans back on. Typically set from 3 minutes to 5 minutes.

- Fans Off
- Liquid Valve Open (energized).
- Suction Valve Open (energized).
- Hot Gas Valve Closed (de-energized)
- Re-Heat Valve Closed (de-energized) If unit is calling for Re-Heat.

Settings

All Air unit settings can be changed by selecting the “Settings” Button located on the left side of each air units Detail Screen. The operator must be logged in to access these screen, the settings are broke down into the following three popup screens once “settings” is selected.



Settings Menu Screen

Temperature Settings

When the "Temp Settings" button is selected the following screen will popup.

AU-2A TEMP & MISC SETTINGS

Temp/Humidity Settings

Temp Setpt: ####

Hi Temp Alarm: ####

Lo Temp Alarm: ####

Humidity Setpt: ####

Unit Enable/Disable

DISABLED

Humidity Override

Temp Offset: ####

Temp Above Set Point

DISABLED

Fan Cycling

Enter 0 in ON time to keep fans off

Fan ON Time: ####

Fan OFF Time: ####

DISABLED

Temperature and Misc Settings Screen

Temp/Humidity Setting Values

These set points are used for operation and alarm settings.

The Temp Set Point value can be set from -40 to 140 Degree's F

The Hi Temp Alarm value can be set from -40 to 140 Degree's F

The Lo Temp Alarm value can be set from -40 to 140 Degree's F

The Humidity value can be set from 0 to 100 %

The Humidity Override value can be set from 3 to 7 Degree's F

Temperature Set Points

The system controller will add one (1) and subtract one (1) from the set point that is entered to give a two (2) degree dead band. When the temperature is above this deadband the controller will put the unit in cooling and will remain in cooling mode until the temperature gets 1 degree below the set point.

Humidity Set Points

The system controller will add one (1) and subtract one (1) from the set point that is entered to give a two (2) % dead band. When the humidity is above this deadband the controller will energize the Re-Heat coil and re-heat will remain on until the humidity gets 1 % below the set point.

Humidity Override can be enabled or disabled with the button shown above. If enabled it will stop the unit Re-Heat from being energized "On" when the area temperature is "Temp Offset" value above the Temp set point. The Reheat will not be allowed back on until the temperature reaches set point.

Unit Enable/Disable

When an Air Unit is Disabled it will go through a 5-minute Pumpout then all solenoids are De-Energized and fans turned off until its Enabled again. The toggle switches in control panels will override this function

Fan Cycling Settings

If Fan Cycling is not enabled the fans will remain on at all times (except defrost). If Fan Cycling is enabled and the Evaporator is satisfied the fans will run for the amount of time entered into the On time and will cycle off for the amount of time entered into the Off time. Upon a call for cooling the fans will be turned on even if currently off from Fan Cycling. To keep fans off when an air unit is satisfied a zero must be entered in the fan cycle on time.

The Fan On Time Value can be from 0 to 120 Minutes

The Fan Off Time Value can be from 1 to 120 Minutes

Defrost Settings

When the "Defrost Settings" button is selected the following screen will popup. The operator can enter the following values for the defrost cycle under Enter Values in Minutes.

The Pump Out Value can be set from 1 to 20 Minutes

The Hot Gas Value can be set from 1 to 60 Minutes

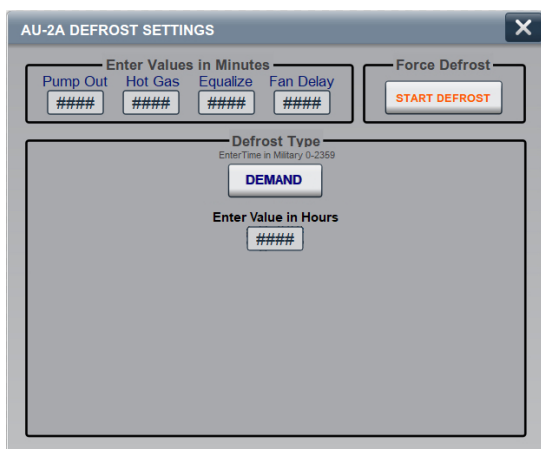
The Bleed Value can be set from 1 to 20 Minutes

The Fan Delay Value can be set from 1 to 20 Minutes

Demand Defrost

The operator can select whether to use Demand or Scheduled Defrost. By selecting the button below Defrost Type the system will toggle between these two modes. If the Demand Defrost is enabled the controller will accumulate the amount of time which has elapsed in the cooling cycle (Liquid Valve On Time). When this accumulated time reaches the Demand Interval Set Point, the unit will begin the defrost cycle. The Demand Defrost accumulated time will be reset upon finishing the defrost cycle.

The Demand Defrost interval Value can be set from 1 to 168 Hours

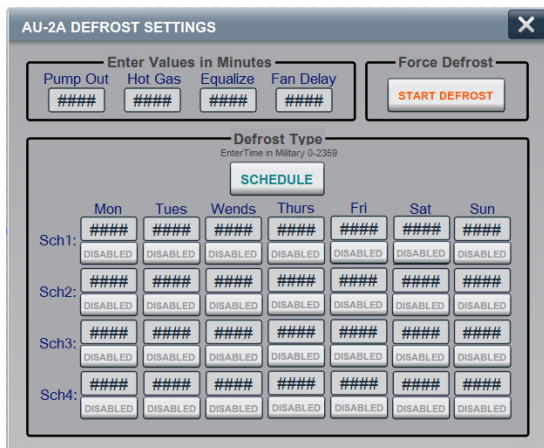


The screenshot shows the 'AU-2A DEFROST SETTINGS' window. At the top, there are four input fields for 'Enter Values in Minutes': 'Pump Out', 'Hot Gas', 'Equalize', and 'Fan Delay', each with a '####' placeholder. To the right is a 'Force Defrost' section with a 'START DEFROST' button. Below these is the 'Defrost Type' section, which has a 'DEMAND' button selected. Underneath is an 'Enter Value in Hours' field with a '####' placeholder.

Demand Defrost Settings

Scheduled Defrost

The Scheduled Defrost function allows a seven day defrost schedule to be entered at the operator interface. In this mode an evaporator can be scheduled to defrost up to four (4) times a day. The Defrost Type button is used to choose between a scheduled defrost or a demand type defrost. Schedules for each day of the week must be enabled and time to defrost entered in Military Time. If all defrost schedules are left disabled the unit will not defrost.



The screenshot shows the 'AU-2A DEFROST SETTINGS' window. The top section is identical to the Demand Defrost settings. In the 'Defrost Type' section, the 'SCHEDULE' button is selected. Below this is a 4x7 grid of settings for four schedules (Sch1, Sch2, Sch3, Sch4) across seven days (Mon, Tues, Wends, Thurs, Fri, Sat, Sun). Each cell contains a '####' placeholder for time and a 'DISABLED' status indicator.

Scheduled Defrost Settings

Force Defrost

The Force Defrost function will allow for an Evaporator to be put into Defrost at any time.

Note: Only one evaporator can go into defrost in at a time so if another is defrosting it will wait until other units are finished.

Defrost Groups

The seven air units controlled from this panel are tied together in one defrost group. Only one of these units can be in defrost at a time.

Temperature Setback

When the "Schedule Settings" button is selected the following screen will popup The Temperature Setback function will enable the Evaporators in each area to be set to a different temperature control point at a specific time for a specified duration. There are seven Setback Schedules available to use and must be enabled "below each schedule" in order for that schedule to be active. Below is a description for Schedule 1's settings.

Temp: Enter the new temperature setpoint desired during the scheduled period.

Day In: Select what day of week you want the schedule to start.

Time In: Enter what time on Day In you want the schedule to start, Military time 0 to 2359

Day Out: Select what day of week you want the schedule to stop.

Time In: Enter what time on Day Out you want the schedule to stop, Military time 0 to 2359

Enable/Disable: button for Schedule 1

	Schedule 1	Schedule 2	Schedule 3	Schedule 4	Schedule 5	Schedule 6	Schedule 7
Temp:	###	###	###	###	###	###	###
Day In:	THURS	THURS	THURS	THURS	THURS	THURS	THURS
Time In:	####	####	####	####	####	####	####
Day Out:	THURS	THURS	THURS	THURS	THURS	THURS	THURS
Time Out:	####	####	####	####	####	####	####
	DISABLED	DISABLED	DISABLED	DISABLED	DISABLED	DISABLED	DISABLED

Setback "Schedule Settings"

Time Delay After Power Failure

Upon a power failure the Evaporators will delay sixty (60) seconds between each evaporator before restarting when power has been returned to the panel.

Manual Control

A Selector Switch "Manual Power Feeds" on the control panel door enables/disables power to the manual side of the Switch Banks located in the Air Unit Control Panel. This Switch should only be in the "ON" position if air units are being controlled manually. The Toggle HOA switch banks enable manual control of each air units valves and fans. These switches must be placed in appropriate position for the desired refrigeration cycle (cooling/defrost). This feature should only be used if the PLC fails or some other fault condition causes the operator to need to override all automatic controls. Normal operation all Switches should be in the Auto position.

Manual Cooling

Switch Bank Positions for Unit Cooling:

- Liquid Valve Switch Bank toggle in HAND (energized).
- Suction Valve Switch Bank toggle in HAND (energized).
- Hot Gas Valve Switch Bank toggle in OFF (de-energized).
- Re-Heat Valve Switch Bank toggle in HAND or OFF.
- Fans Switch Bank toggle in HAND (fans running).

Switch Bank Positions for Unit Satisfied:

- Liquid Valve Switch Bank toggle in OFF (de-energized).
- Suction Valve Switch Bank toggle in HAND (energized).
- Hot Gas Valve Switch Bank toggle in OFF (de-energized).
- Re-Heat Valve Switch Bank toggle in HAND or OFF.
- Fans Switch Bank toggle in HAND or OFF.

Manual Defrost

Pump Out Cycle:

Pumpout will remove any existing ammonia from the coil and is typically set from 5 minutes to 30 minutes

- Liquid Valve Switch Bank toggle in OFF (de-energized).
- Suction Valve Switch Bank toggle in HAND (energized).
- Hot Gas Valve Switch Bank toggle in OFF (de-energized).
- Re-Heat Valve Switch Bank toggle in OFF (de-energized).
- Fans Switch Bank toggle in HAND (fans running).

Hot Gas Cycle:

Removes any frost/ice that has formed on the coil and is typically set from 10 minutes to 30 minutes.

- Liquid Valve Switch Bank toggle in OFF (de-energized).
- Suction Valve Switch Bank toggle in OFF (de-energized).
- Hot Gas Valve Switch Bank toggle in HAND (energized).
- Re-Heat Valve Switch Bank toggle in OFF (de-energized).
- Fans Switch Bank toggle in OFF (fans off).

Equalize/Bleed Cycle:

Brings the coil back down from Discharge (hot gas) pressure to suction pressure and is typically set from 3 minutes to 10 minutes.

- Liquid Valve Switch Bank toggle in OFF (de-energized).
- Suction Valve Switch Bank toggle in OFF (de-energized).
- Hot Gas Valve Switch Bank toggle in OFF (de-energized).
- Re-Heat Valve Switch Bank toggle in OFF (de-energized).
- Fans Switch Bank toggle in OFF (fans off).

Fan Delay Cycle Mode:

Chills the coil back down and freezes any water droplets on the coil before turning the fans back on. Typically set from 3 minutes to 5 minutes.

- Liquid Valve Switch Bank toggle in ON (energized).
- Suction Valve Switch Bank toggle in ON (energized).
- Hot Gas Valve Switch Bank toggle in OFF (de-energized).
- Re-Heat Valve Switch Bank toggle in OFF (de-energized).
- Fans Switch Bank toggle in OFF (fans off).

Alarms\Warning

Operator adjustable High Temperature Alarm set point.

Operator adjustable Low Temperature Alarm set point.

Fan Fail Start Alarm, fan told to start and no auxiliary conformation back from the starter.

Temperature Sensor Failed Over/Under Range Alarm.

Humidity Sensor Failed Over/Under Range Alarm.

Unit Disabled Warning.

Parts List Air Unit Control Panel

See prints for Item No. locations

Item No.	Qty.	Description	Part Number	Manufacturer
1	1	5069 Compact I/O 16 channels AC input modules, supporting both 120 & 240 VAC signals	5069-IA16	Allen Bradley
2	2	5069 Compact I/O 8 Channel Voltage/Current Analog Input Module	5069-IF8	Allen Bradley
3	1	Controller, CompactLogix 5380, 600 KB User Memory, 8 I/Os, 16 EtherNet/IP Devices	5069-L306ER	Allen Bradley
4	4	5069 Compact I/O 16 channels non isolated normally open relay output module.	5069-OW16	Allen Bradley
5	7	5069 Compact I/O 18 pins Spring type terminal block kit in a pack of 1 pcs	5069-RTB18-SPRING	Allen Bradley
6	1	5069 Compact I/O Power terminal RTB kit for both 4 and 6 pin Spring type	5069-RTB64-SPRING	Allen Bradley
7	1	5069 Compact I/O 6 pin Spring type RTB	5069-RTB6-SPRING	Allen Bradley
8	1	Stratix 2000 Switch, Unmanaged, 5 Copper Ports	1783-US5T	Allen-Bradley
9	1	5069 Compact I/O Field Potential Distributor	5069-FPD	Allen-Bradley
10	1	UL 489, Miniature Circuit Breaker, 2 Pole, C Curve, 15 Amp	FAZ-C15/2-NA	Eaton
11	10	UL 1077, Miniature Circuit Breaker, 1 Pole, C Curve, 2 Amp	FAZ-C2/1-SP	Eaton
12	2	UL 489, Miniature Circuit Breaker, 1 Pole, C Curve, 20 Amp	FAZ-C20/1-NA-SP	Eaton
13	6	UL 1077, Miniature Circuit Breaker, 1 Pole, C Curve, 5 Amp	FAZ-C5/1-SP	Eaton
14	2	UL 489, Miniature Circuit Breaker, 3 Pole, C Curve, 6 Amp	FAZ-C6/3-NA	Eaton
15	5	Power Distribution Block, 80A, 1 Input, 6 Output, 1 Pole, DIN Rail, UD Series	UD-80A (569010)	ERIFLEX
16	16	Fuse, 2 Amp, Glass for (2002-1611)	LTFS-218002P	Little Fuse
17	18	Wire Duct Cover 3"	C3LG6-F	Panduit
18	42	Wire Duct Cover 4"	C4LG6-F	Panduit
19	18	Wire Duct, 3" x 4" p/ft	F3X4LG6	Panduit
20	42	Wire Duct, 4" x 4" p/ft	F4X4LG6	Panduit
21	1	Ground Block for 14-2 Awg Wire	LAMA2-14-QY	Panduit
22	1	Temperature Switch	3110000	Rittal
23	1	PowerPact B Circuit Breaker, 60A, 3P, 600Y/347V AC, 14kA at 600Y/347 UL EverLink	BDL36060	Schneider
24	2	TeSys D control relay - 5 NO - <= 690 V - 110 V AC standard coil	CAD50F7	Schneider
25	2	TeSys GV, top feed terminal block, for busbars, 3 P, for TeSys D and TeSys U	GV1G09	Schneider
26	1	Linergy FT - Comb busbar - 63 A - 3 tap-offs - 45 mm pitch	GV2G345	Schneider

27	1	Linergy FT - Comb busbar - 63 A - 4 tap-offs - 45 mm pitch	GV2G445	Schneider
28	4	TeSys GV2 Manual Starter and Protector, thermal magnetic circuit protector, rotary handle, 4.0 to 6.3 A, screw clamp	GV2P10	Schneider
29	3	TeSys GV2 Manual Starter and Protector, thermal magnetic circuit protector, rotary handle, 6 to 10 A, screw clamp	GV2P14	Schneider
30	7	TeSys GV2 and GV3 Manual Starter and Protector, auxiliary contact block, 1 NO and 1 NC, top mount, screw clamp	GVAE11	Schneider
31	2	Safety Cover	LAD9ET1S	Schneider
32	7	TeSys D IEC contactor, 12 A, 3 P, 5 HP at 480 VAC, nonreversing, 110 VAC 50/60 Hz coil	LC1D12F7	Schneider
33	1	3 KVA 460/230 - 230/115 Transformer	3S1F	Square D
34	1	20 Pole Ground Block	PK20GTA	Square D
35	25	TOPJOB*S feedthrough terminal block; rail mount; 4-conductor; 6.2 mm wide; gray	2002-1401	WAGO
36	31	TOPJOB*S feedthrough terminal block; rail mount; 4-conductor; 6.2 mm wide; red	2002-1403	WAGO
37	20	TOPJOB*S feedthrough terminal block; rail mount; 4-conductor; 5.2 mm wide; blue	2002-1404	WAGO
38	10	TOPJOB*S ground terminal block; rail mount; 4-conductor; 5.2 mm wide; green-yellow	2002-1407	WAGO
39	3	End and intermediate plate; 1 mm thick; orange	2002-1492	WAGO
40	16	TOPJOB*S fuse disconnect terminal block; with pivoting fuse holder for 5 x 20 mm fuse; rail mount; 2-conductor; 6.2 mm wide; gray	2002-1611	WAGO
41	1	TOPJOB*S end/intermediate plate; for 2002 series fuse blocks; 2.0 mm wide; orange	2002-1692	WAGO
42	10	TOPJOB*S jumper; for 2002 series; insulated; 10-way; light-gray	2002-410	WAGO
43	3	Steel carrier rail; 35 x 7.5 mm; 1 mm thick; 2 m long; slotted; according to EN 60715; "Hole width 25 mm	210-112	WAGO
44	36	Screwless end stop; 6 mm wide; for DIN-rail 35 x 15 and 35 x 7.5; gray	249-116	WAGO
45	3	End and intermediate plate; 1 mm thick; for quadruple-deck terminal blocks; orange	280-323	WAGO
46	35	4-conductor sensor terminal block; with ground contact; with colored conductor entries; 2.5 mm ² ; CAGE CLAMP*; 2,50 mm ² ; gray	280-570	WAGO
47	2	Interface module; RJ-45; IDC technology; Cat. 6; in mounting carrier; with shield connection	289-195	WAGO
48	8	1-conductor female plug; 100% protected against mismatching; 2.5 mm ² ; Pin spacing 5 mm; 8-pole; 2,50 mm ² ; light gray	721-108/026-000	WAGO
49	3	1-conductor male connector; 100% protected against mismatching; 2.5 mm ² ; Pin spacing 5 mm; 2-pole; 2,50 mm ² ; light gray	721-602	WAGO
50	2	EPSITRON* ECO power supply; three-phase; output voltage 24 VDC; 10 A	787-740	WAGO
51	1	Redundancy Module; 2 x 9 ... 54 VDC input voltage; 2 x 12.5 A input current; 9-54 VDC output voltage; 25 A output current	787-783	WAGO
52	3	16 Port HOA Switch Board	PRO2962	WAGO

Item No.	Qty.	Description	Part Number	Manufacturer
1	1	PanelView 5310, 10 in. Wide Display, Touch screen, Color, 24V DC, Single Ethernet port	2713P-T10CD1	Allen Bradley
2	1	Convenience Receptacle	P-R2-K3RFO-U601	Grace
3	1	Complete fan unit for tool-free mounting on surfaces. Air throughput from 20 - 900 m³/h, protection category IP 54 as standard.	3241110	Rittal
4	1	Outlet Filters Standard	3241200	Rittal
5	1	TSFM716320XC TS 8 Flange Mount Disconnect Floor Mount Enclosure	8018814	Rittal
6	1	35mm Hole Seal	OHS050C	Rittal
7	1	Handle Mechanism	9422A1	Schneider
8	1	B Frame 60" Flex Cable,	9422CSB50	Schneider
9	1	Push Button Spring Return, Recessed, Black	ZB5AA26	Schneider
10	1	Selector switch head Ø22 2-position stay put Ronis 455	ZB5AG2	Schneider
11	1	Green Illuminated selector switch head Ø22 2-position stay put	ZB5AK1233	Schneider
12	3	Harmony, 22mm Push Button, XB5A operators, mounting collar for electrical blocks	ZB5AZ009	Schneider
13	3	Contact Block, Spring Clamp, Normally Open	ZBE1015	Schneider
14	1	Light Block, Spring Clamp Terminals, 110-120V, Green	ZBVG35	Schneider
15	1	WERMA audible alarm buzzer, piezoelectric, 80 dB at 1m, 2.8 kHz tone frequency, continuous tone(s), 115 VAC, permanent LED indicator(s), red, 22mm, panel mount, polycarbonate housing, red, IP65.	15010067	WERMA